Exploratory Data Analysis and Prediction of Air Quality Index Using Weather Variables

Objective

The main aim of this project is to understand how temperature, humidity and wind speed affect air quality (AQI). Using this data, we build a simple model to predict AQI.

Dataset Description

The dataset used in this project contains daily weather data for the month of March. It includes temperature (in Fahrenheit), humidity, wind speed (in mph) and AQI values. The data was collected manually and used to analyze patterns and realationships between these variables.

Data Analysis:

Correlation Analysis

Table 1: Correlation Matrix

	Temp(F)	Humidity	wind speed	AQI
Temp(F)	1.000000	-0.796936	-0.042800	0.133631
Humidity	-0.796936	1.000000	-0.261464	0.189241
wind speed	-0.042800	-0.261464	1.000000	-0.139722
AQI	0.133631	0.189241	-0.139722	1.000000

The correlation matrix shows the relationships between temeprature, humidity, wind speed and AQI. It provides an intial understanding of how these variables are associated with air quality levels, although the relationships appear to be weak.

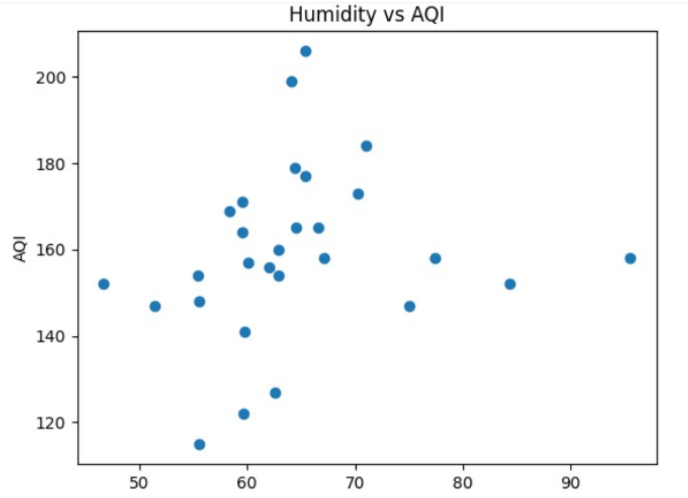
Temperature and AQI show a weak positive correlation (+0.13), indicating that temeprature has a slight impact on AQI but is not a strong factor.

Humidity also shows the weak positive correlation (+0.18), suggesting a minor influence on AQI.

Wind speed has a weak negetive correlation (-0.13), which means high wind speed may help reduce AQI by dispersing pollutants. Overall, no single factor strongly affects AQI, and it depends on a combination of multiple factors. This indicates that AQI is influenced by multiple environmental factors rather than a single dominant variable.

Graph Analysis

Humidity Vs AQI

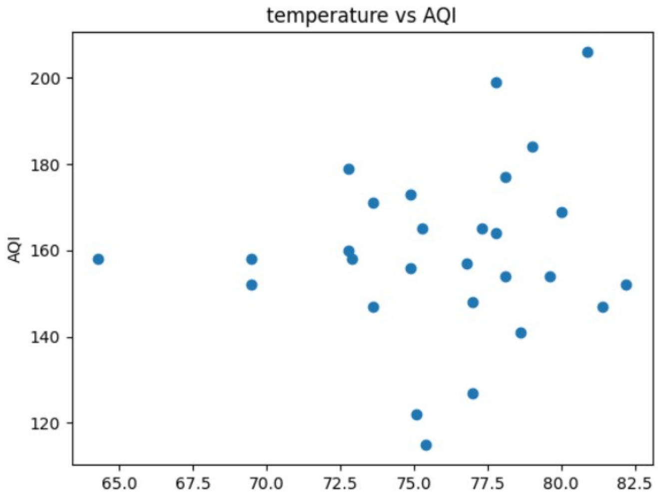


A scatter plot was used to visualize the realtionship between humidity and AQI. The data points are mostly concentrated in the humidity range of 60-70%, where AQI values lie between approximately 140 and 180. This indicates a slight upward trend, suggesting a weak positive relationship between humidity and AQI.

However, the data points are somewhat scattered, indicating that realtionship is not strong. This supports the correlation results, which show that humidity has only a minor influence on AQI.

Higher humidity may allow pollutants to absorb moisture, increasing their size and concentration in the air, which can contribute to higher AQI levels.

Temperature VS AQI

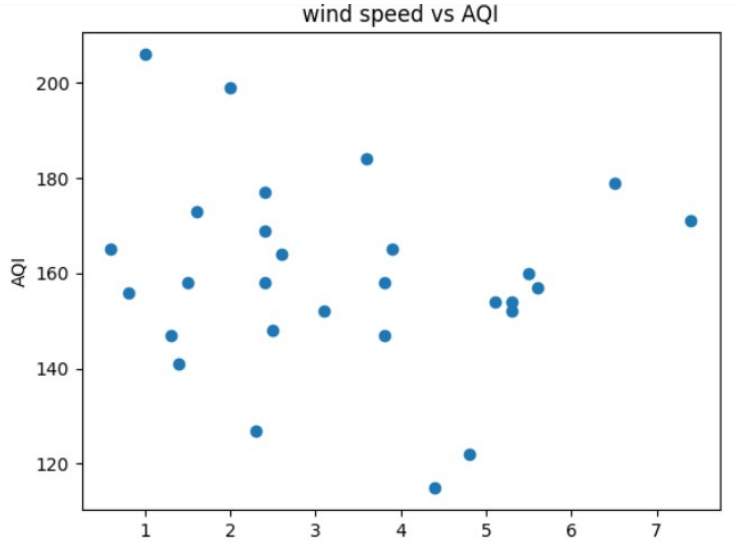


The scatter plot shows that the data points are widely scattered, indicating that temperature does not have a strong or direct effect on AQI. There is no clear pattern, which suggests a weak realtionship between the two vairables.

However, temperature can still influence air quality indirectly. Higher temperature can cause warm air to rise, leading to vertical mixing, which may help disperse pollutants and improve AQI. On the other hand, high temperatures can also increase chemical reactions involving pollutants such as nitrogen oxides (Nox) and volatile organic compounds (VOCs), which may lead to the formation of additional pollutants.

Overall, temperature has a limited and indirect impact on AQI, and air quality is influenced by multiple environmental factors.

Wind vs AQI



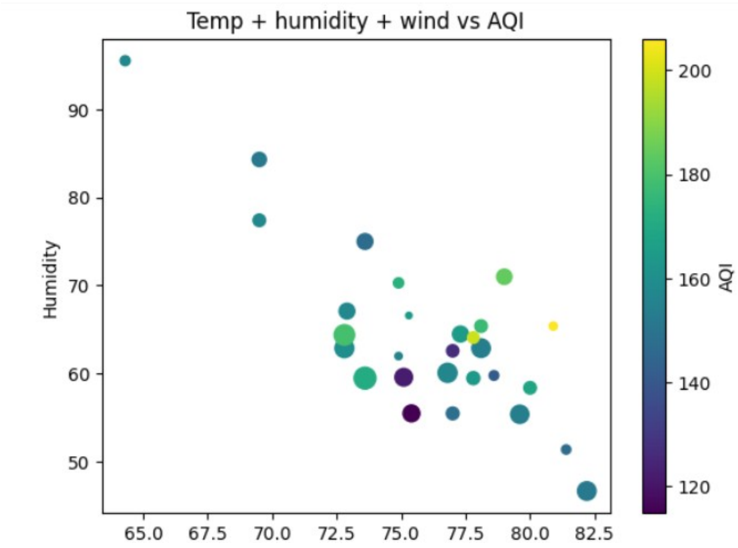
The scatter plot shows a high level of varation in AQI across different wind speeds, with no clear or consistant pattern. While some data points at lower wind speed (1-2 mph) show AQI between 140-180, there are also higher values observed, indicating inconsistency.

At moderate wind speeds (around 4-5 mph), a few lower AQI values are observed, which may suggest that wind can help disperse pollutants. However, at higher wind speed (around 6-7 mph), AQI values are again relatively high in some cases.

This mixed pattern indicates that wind speed alone does not have a strong or consistent effect on AQI. The limited dataset (only 28 days) may also contribute to this variability. Additionally, wind can sometimes carry pollutants from other areas, which may increase AQI, but this cannot be confirmed without additonal data.

Overall, wind speed has a complex and limited influence on AQI.

Temperature + Humidity + Wind speed vs AQI



The combined scatter plot shows the relationship between temperature, humidity, wind speed, and AQI. Temperature is represented on the x-axis, humidity on the y-axis, and wind speed is indicated by the size of the circles, while color represents AQI levels.

The data points are still widely scattered, indicating that there is no strong or clear pattern when all variables are considered together. However, a slight tendency can be observed where moderate humidity levels (around 50–60%) and higher wind speeds (larger circles) are associated with relatively lower AQI values (around 120–160).

This suggests that wind speed may help in reducing AQI by dispersing pollutants, while temperature and humidity contribute indirectly. However, the relationship is not consistent across all data points, and the limited dataset size may affect the reliability of this observation.

Overall, AQI appears to be influenced by a combination of multiple factors rather than a single dominant variable.

An additional observation can be seen in the extreme data points. At a lower temperature (~65F) with very high humidity (above 90%) and low wind speed, AQI is relatively high (around 200). In contrast, at a higher temperature (~82.5 F) with moderate humidity (~50%) and higher wind speed, AQI is lower (around 120).

This comparison suggests that wind speed and humidity may have a stronger combined influence on AQI than temperature alone. However, since this is based on only a few data points, it should be interpreted with caution.

Model Building and Prediction

A linear regression model was developed to predict AQI using temperature, humidity, and wind speed as input variables. The dataset was split into training and testing sets to evaluate model performance. The model achieved a Mean Absolute Error (MAE) of approximately 11, indicating that the predicted AQI values differ from the actual values by about 11 units on average.

The model performs reasonably well for moderate AQI values but shows lower accuracy for extreme cases. This suggests that while the model captures general trends in the data, it struggles with outliers. The limited dataset size and weak correlations between the variables may contribute to this reduced accuracy. Overall, the model provides a basic understanding of how environmental factors influence AQI, but more data and additional features such as pollutant concentrations (e.g., PM2.5, NO_2) would be required for more accurate predictions.

Limitation of the study

This study has several limitations that may affect the accuracy and generalization of the results. Firstly, the dataset is relatively small, consisting of limited daily observations, which reduces the reliability of the analysis and model predictions.

Secondly, the model uses only a few weather variables (temperature, humidity, and wind speed), while AQI is influenced by many other important factors such as pollutant concentrations (PM2.5, PM10, NO_2), traffic density, industrial emissions, and seasonal variations.

Additionally, the relationships observed between variables are weak, as indicated by the correlation analysis and scattered plots, which limits the predictive power of the model.

Finally, the linear regression model assumes a linear relationship between variables, which may not fully capture the complex and non-linear nature of environmental factors affecting air quality.

Conclusion

This study analyzed the relationship between weather parameters—temperature, humidity, and wind speed—and Air Quality Index (AQI). The correlation analysis and graphical visualizations revealed that these variables have only weak relationships with AQI, indicating that no single factor strongly influences air quality on its own.

The scatter plots showed that the data points are widely distributed, suggesting high variability and the presence of multiple influencing factors. While slight trends were observed, such as higher wind speeds being associated with relatively lower AQI in some cases, these patterns were not consistent across all observations.

The linear regression model provided a basic predictive framework, achieving a Mean Absolute Error (MAE) of approximately 11. Although the model performed reasonably well for moderate AQI values, it was less accurate for extreme values, highlighting the limitations of using a simple linear approach on a small dataset.

Overall, the findings suggest that AQI is influenced by a combination of multiple environmental and external factors rather than a single dominant variable. For more accurate analysis and prediction, future work should include larger datasets and additional variables such as pollutant concentrations and human activity indicators.